STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Documentation

Submitted by: Lavanya V

Department: Computer Science and Engineering

College: Vemana Institute of Technology

Course: Python with AI – EWB Courses

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1. Project Title

Student Performance Analytics System

The Student Performance Analytics System is a Python-based data analysis project developed to analyze student marks and attendance. It uses Python, NumPy, and Pandas to calculate academic performance and generate meaningful results from student records.

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2. Objective

The main objective of this project is to analyze student academic performance using Python programming and data analysis libraries.

Calculating marks, averages, grades, and pass/fail results manually for multiple students can be time-consuming. This project simplifies the process by automatically calculating student performance statistics using data stored in a CSV file.

The system helps identify top-performing students, compare subject-wise performance, calculate the overall class average, and analyze student attendance. It provides clear and organized results that make student performance easier to understand.

3. Technologies Used

- **Python:** Used to write the program and perform calculations using conditions, loops, and functions.
- **NumPy:** Used for numerical calculations, including finding the overall class average.
- **Pandas:** Used to read the CSV dataset, organize student records, calculate totals and averages, and analyze performance.
- **CSV:** Used to store student details, subject marks, and attendance.
- **Visual Studio Code:** Used to write and execute the Python program.
- **GitHub:** Used to store and share the project files.

4. Dataset Description

The project uses a CSV dataset named student_marks_attendance_20.csv, containing 20 student records.

The dataset includes student information, department, marks in three subjects, and attendance percentage. This information is processed to calculate academic performance.

Column Name Description

Student ID	Unique identification number of each student
Name	Name of the student
Department	Department to which the student belongs
Math	Marks obtained in Mathematics
Physics	Marks obtained in Physics
Chemistry	Marks obtained in Chemistry
Attendance (%)	Attendance percentage of the student

The dataset is loaded into a Pandas DataFrame for processing and analysis. The calculated results are saved separately in student_performance_results.csv.

5. Implementation

The program first reads the student dataset from the CSV file using the Pandas read_csv() function and stores the records in a DataFrame. It displays the student details and the total number of student records.

The program calculates the total marks obtained by each student in Mathematics, Physics, and Chemistry. It then calculates the average marks to understand each student's academic performance.

A function named calculate_grade() assigns grades based on average marks. Students scoring 90 and above receive grade A, those scoring from 75 to below 90 receive grade B, those scoring from 60 to below 75 receive grade C, those scoring from 40 to below 60 receive grade D, and students scoring below 40 receive grade F.

Another function named check_result() determines whether a student has passed or failed. A student passes if they score at least 35 marks in every subject. If the student scores below 35 in any subject, the result is marked as Fail. A loop is used to check the marks in all three subjects.

NumPy and Pandas operations are used to calculate the overall class average, identify the highest and lowest student averages, and find the top-performing student. The program also calculates subject-wise average marks to identify the best-performing and lowest-performing subjects.

The attendance analysis calculates the average attendance percentage and identifies students whose attendance is below 75%. Finally, the processed student data, including total marks, averages, grades, and pass/fail results, is saved in a separate CSV file named student_performance_results.csv.

The implementation uses Python variables, data types, loops, conditional statements, functions, NumPy calculations, Pandas DataFrames, and CSV file handling.

6. Key Features

- Reads student records from a CSV dataset.
- Calculates total and average marks for each student.
- Assigns grades based on average marks.

- Identifies students as Pass or Fail.
- Calculates the overall class average.
- Finds the highest and lowest student averages.
- Identifies the top-performing student.
- Calculates subject-wise average marks.
- Identifies the best-performing and lowest-performing subjects.
- Calculates average student attendance.
- Displays students with attendance below 75%.
- Saves the analyzed results in a separate CSV file.

7. Output

When the program is executed, it displays the student details and the results of the analysis in the terminal.

The output includes the total number of students, total and average marks, grades, pass/fail results, overall class average, highest and lowest student averages, top-performing student, pass/fail count, subject-wise averages, and attendance analysis.

The program also generates student_performance_results.csv, which contains the original student records along with the calculated performance details.

8. Final Outcome

The Student Performance Analytics System demonstrates how Python, NumPy, and Pandas can be used to analyze student academic performance.

The program processes 20 student records and calculates total marks, average marks, grades, and pass/fail results. It also identifies top-performing students, compares subject-wise averages, and analyzes attendance.

The final results are displayed clearly and saved in a separate CSV file for future reference. The project reduces manual calculations and demonstrates the practical application of Python programming and data analysis concepts.

9. Challenges & Learning

One challenge was ensuring that the column names used in the Python program matched the names in the CSV dataset. This was handled by using the correct column names in the code.

Another important part was applying the pass/fail condition correctly. The program checks each subject separately to ensure that a student who scores below 35 in any subject is marked as Fail.

Through this project, I learned how to read CSV files using Pandas, work with DataFrames, perform numerical calculations using NumPy, create reusable functions, and apply loops and conditional statements. I also learned how to filter student records, analyze academic performance, calculate attendance statistics, and save processed data into a CSV file.

This project helped me understand how the Python concepts taught in class can be applied to a practical data analysis problem.

10. GitHub Repository

GitHub Repository Link: https://github.com/lavanyavreddy303-cell/Student-Performance-Analytics-System

The GitHub repository contains the project files, including the Python program, student dataset, and README file. It provides a central location to access and review the project implementation.

Conclusion

The Student Performance Analytics System is a simple and useful Python-based data analysis project. It demonstrates the use of Python fundamentals, NumPy, and Pandas to process student data and generate meaningful academic performance statistics.

The project helped me improve my understanding of data handling, functions, loops, conditional statements, and numerical analysis. It also provided practical experience in working with CSV datasets and organizing a project for GitHub submission.